

Capacity is an interval: the traffic rule sets the floor, the network sets the ceiling

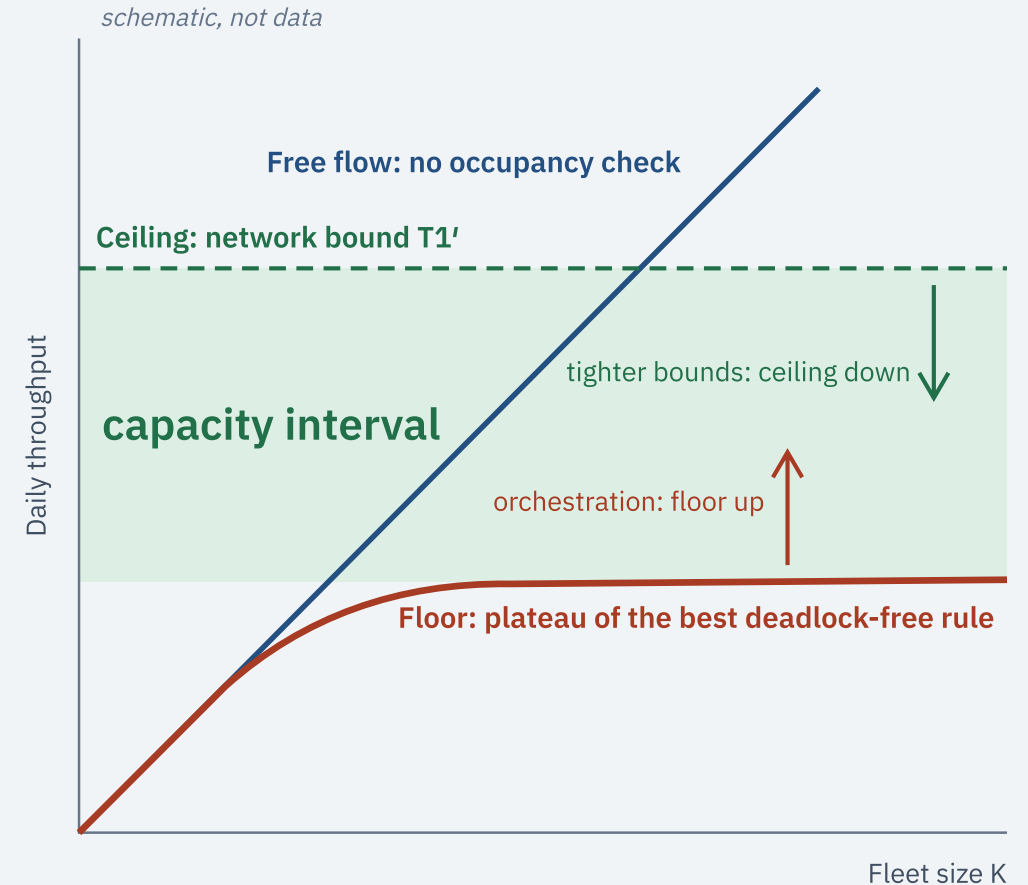
The flag experiment on two real yards: the capacity interval, the fleet–throughput trapezoid, the network-bound regime and the free-flow error (C1, C2, H1, H3)

When transporters own the road: capacity bounds and operating regimes in shipyard block logistics (working title)

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Data: thesis-core commit 83c16bf (T17–T21, all reviewed)



The sub-study in one minute

1

Question

When loaded vehicles own the road, how much can block transport deliver, when do more vehicles stop helping, and how wrong is free flow?

2

Method

Two real networks, Okpo and Yantai; three traffic rules; fleets from 1–5 up to 150 vehicles, 10 seeds each; a linear program gives the network bound $T1'$.

3

Results so far

Capacity intervals: Okpo [627, 1,720], Yantai [455, 1,252], the floor only 36% of the ceiling on both; network-bound from $K^* \approx 50$ (Okpo) and ≈ 25 (Yantai); the service-time gap is superlinear, $\alpha = 1.22\text{--}1.38$.

4

Next

Task volume moves to whole-yard scale (300–600 a day); the preview puts Okpo's vehicles needed past K^* , while Yantai's 23–46 a day leave free flow sufficient. T22 re-runs this and draws the work-zone chart.

Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.

This sub-study answers four items of the research manifesto: C1, C2, H1 and H3

Manifesto wording (v1, 9 Oct 2026)		What this sub-study delivers	
C1	Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound is set by the blocking-time structure and can be computed by linear programming. Attainable capacity lies within the interval [best deadlock-free rule, network bound]		Both yards' capacity intervals and the signature figure; case-by-case check of T1'
C2	Throughput first rises linearly with fleet size, then enters a plateau set by the network; when waiting occupies the road, it falls beyond a jamming threshold. The free-flow model sees neither of the last two parts		First two parts: both yards' flag curves; the conditions for the fall belong to H2 (T22 no-clearing switch, T26 holding area)
H1	At a reasonable daily task volume and a realistic fleet size, the system is in the network-bound regime, at least at erection peaks or near the dock mouth		The manifesto's criteria and three-part form: K^* , vehicles needed, the density criterion; the work-zone chart
H3	The free-flow model's underestimate of completion time and of vehicles needed grows superlinearly with vehicle density		The exponent α of the service-time gap $\Delta(K)$; the ratio of vehicles needed at whole-yard volume

The flag experiment: two real yard layouts, three traffic rules, one task stream

Free flow	No occupancy check; vehicles never interact (the field's default)
Whole-route reservation	Books the whole route before departing loaded, leaves just in time; deadlock-free. The best such rule so far: the interval's floor
Segment request	One segment at a time; on deadlock a vehicle leaves the network for 900 s (teleport clearing) Reference rule (teleport clearing): supports no C or H on its own

- **Demand:** saturated (a task whenever a vehicle is free, so it measures capacity); for comparison Okpo 97 / 194 a 16-h day (4-transporter subset, Yim 2008), Yantai 23 / 46 (from public steel use)
- **Fleets and seeds:** K = 5–40 (Yantai 1–40) in steps of 1, 45–150 in steps of 5; 10 seeds per point; 16-h days, 4 days simulated, day 1 warm-up

	OKPO	YANTAI
Roads (count / km)	721 / 40.3	140 / 8.6
Junction resources	334	58
Stops	421	140
Comparison volume: regular / peak	97 / 194	23 / 46
Where T1' binds (seeds)	dock road 161 5/10, road 010-6 4/10	the two doors of building 014 9/10

- **Ceiling T1':** an occupancy LP with width bin-packing rows (Fekete–Schepers), exact rationals; checked case by case, none exceeds it (17,760 occupancy runs on the two yards in T21)
- Maps and granularity:** Hanwha Ocean Okpo rev 1824 (network identical to rev 1825, which adds crane spans), CIMC Raffles Yantai rev 395 (identical to rev 396), traced by the author on public imagery; neither imagery nor geometry is shown. Each road is one resource (Okpo's dock road 161 included); Yantai's doors are held exclusively while loading.

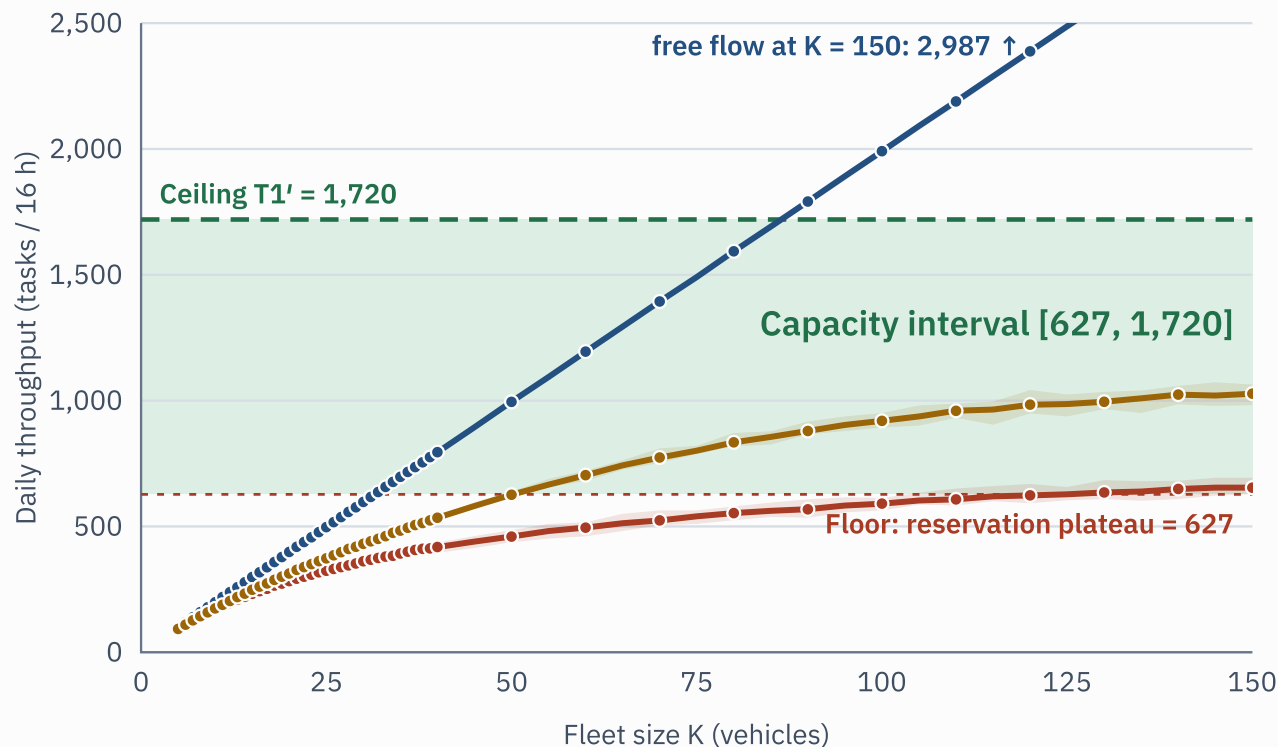
Sources: thesis-core 83c16bf, docs/T17_flag_report.md §1–2, T18_report.md, T20_report.md, T21_report.md §0–1; config.json of results/a0ab16fceb94, 49d2383872e7 (Okpo) and b82aa19b22af (Yantai). Sources of speeds, platform size and handling times: T17 report §2.

The capacity interval: the reservation plateau below, T1' above, and on both yards the floor is only 36% of the ceiling

Okpo

Yantai

Saturated demand (capacity)



Free flow Whole-route reservation Segment request (reference rule, teleport clearing)

Capacity interval Ceiling $T1'$ (bounds occupancy models only)

[627, 1,720]

Capacity interval (a day): the floor is the reservation plateau (mean of $K = 100$ – 150), the ceiling $T1'$ (mean of 10 seeds, 1,436–1,948 across seeds)

36%

The floor reaches only this share of the ceiling: the gap is the traffic rule, which orchestration aims to close

984 57%

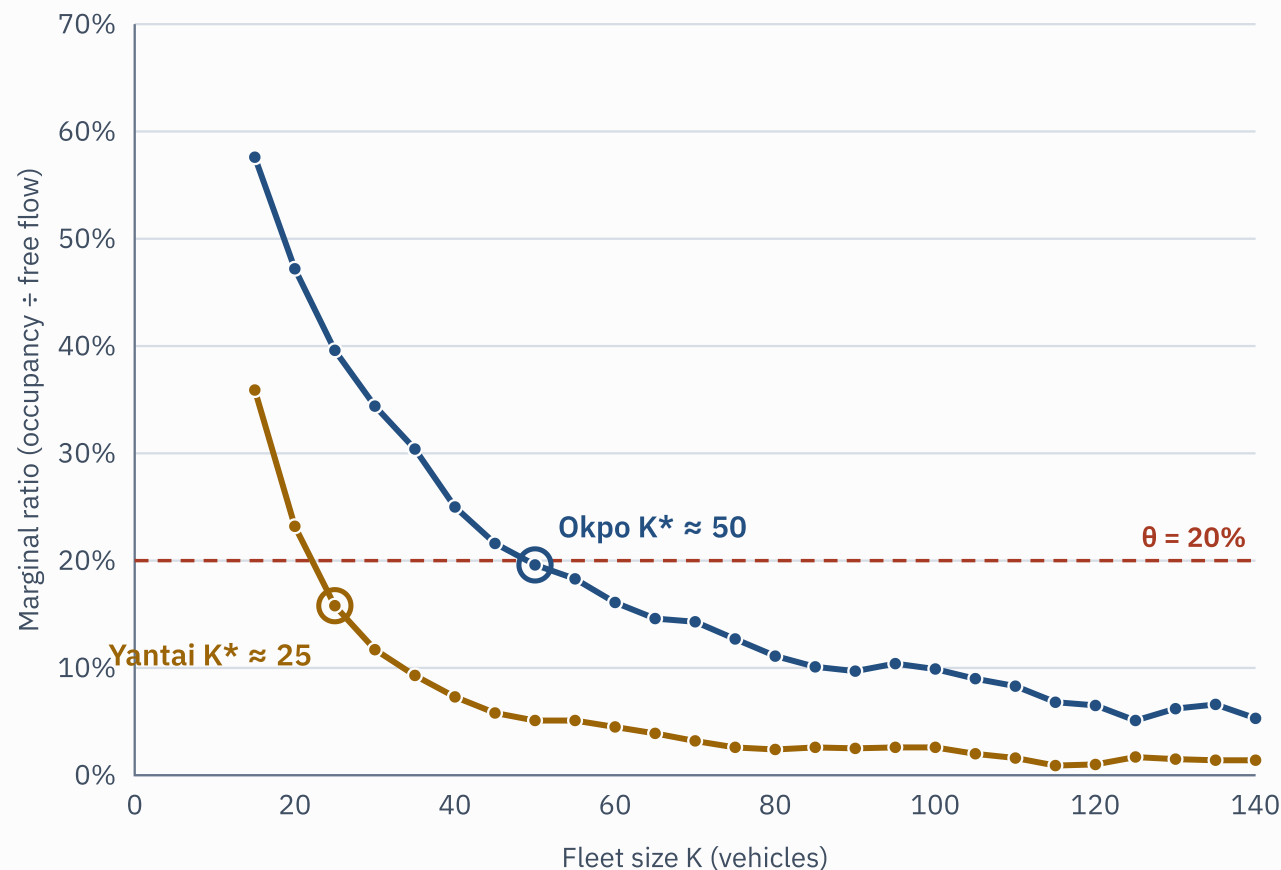
Segment-request plateau (reference rule with teleport clearing, for reference only)

$K \approx 50$

Network-bound from $K^*(20\%)$, range 50–60

Sources: thesis-core 83c16bf; Okpo results/a0ab16fceb94 and 49d2383872e7, Yantai b82aa19b22af; plateaus are means over $K = 100$ – 150 (t20_summary.json), $T1'$ the mean of 10 seeds. Curves are means of 10 seeds, shading min–max. Conditions: saturated demand, each road one resource; Okpo rev 1824, Yantai rev 395. $T1'$ bounds only the occupancy models; free flow checks no occupancy and may cross it.

H1 criterion $K^*(20\%)$: from $K \approx 50$ on Okpo and $K \approx 25$ on Yantai, one more vehicle adds less than 20% of its free-flow gain



■ Okpo ■ Yantai — $\theta = 20\%$

$K \approx 50$

Okpo: network-bound from here (range 50–60)

$K \approx 25$

Yantai: a small network starts much earlier

58% → 34%

Okpo, $K = 15 \rightarrow 30$: one more vehicle's gain as a share of free flow; usual fleet sizes are already in the transition

Manifesto criterion: $K^*(20\%)$ is where the marginal throughput of one more vehicle first falls below 20% of free flow and stays below thereafter.

Sources: thesis-core 83c16bf, t18_summary.json (Okpo) and t20_summary.json (Yantai rev 395). Marginal ratio = occupancy gain ÷ free-flow gain over a 20-vehicle window centred on K; the range is read from the ratio's 95% interval; at $\theta = 10\%$: Okpo 100, Yantai 35. Conditions: whole-route reservation, saturated demand, each road one resource.

H1 in the manifesto's three parts: part 1 has readings, part 2 awaits whole-yard volumes, part 3 awaits physical rules

1

The capacity interval and the start of the network-bound regime K^*

Okpo [627, 1,720], $K^* \approx 50$; Yantai [455, 1,252], $K^* \approx 25$ (whole-route reservation, saturated demand)

readings in hand

T18–T20; floor candidate
segment_safe in T22

2

At whole-yard volume, where the vehicles needed sit relative to K^* , and the gap to free flow

Preview from the saturated curves: at 600 a day Okpo needs 101–105 vehicles under reservation against 30 in free flow; Yantai's 23–46 a day need 1–2 under every rule

preview; T22 re-runs

finite volumes, 16 h and 24 h, one
caliber for both yards

3

Whether the dock mouth becomes the yard's bottleneck depends on whether vehicles waiting for the crane block the way to other stops: where they wait, and whether the dock road is used in sections

The steep fall of the yard plateau appears so far only under segment request (reference rule, teleport clearing) with road 161 as one resource; under reservation vehicles wait off the road and the plateau does not move with crane utilisation

T22 running

Imagery check by the author:
neighbouring sites stay reachable →
Okpo main scenario becomes
segmented blocking, whole road as
comparison

Vehicles-needed criterion

Vehicles needed (mean throughput $\geq 99\% \times$ daily volume) $\geq K^*$ means network-bound

Density criterion

Daily volume / network bound; volume at whole-yard caliber, converted the same way for both yards

Conclusive either way

Holds: shipyard transport has network-bound capacity. Not: a density criterion bounds where free flow suffices

Task volume at whole-yard scale: 97 / 194 is a four-transporter subset; large Korean yards move 300–600 blocks a day

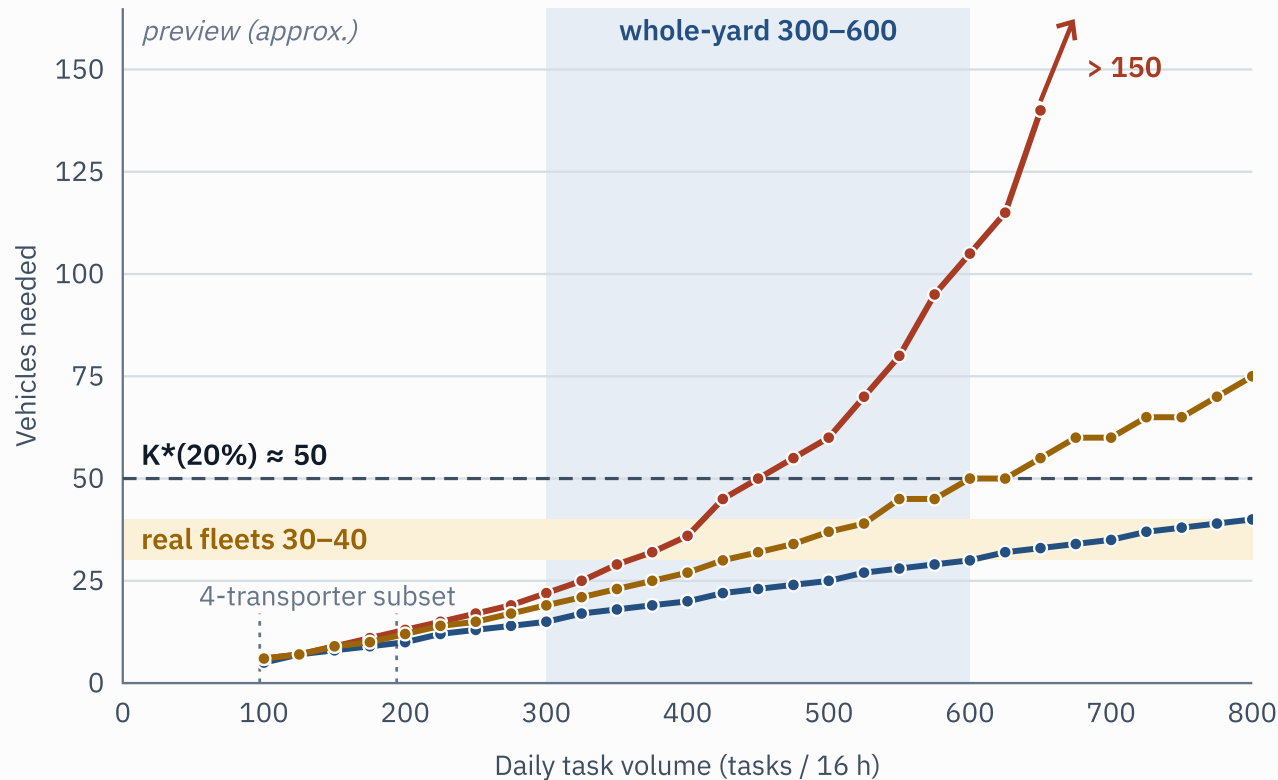
- **Where 97 comes from:** Roh & Cha's 97 blocks come from Yim et al. 2008: one day, 8 a.m. to midnight, in one large yard, moved by 4 transporters. The same paper says a large yard moves over 300 blocks a day
 - **Another large Korean yard:** about 600 a day with about 30 transporters (Shen et al. 2018; the yard is not named)
 - **Okpo itself:** converted as for Yantai (steel ÷ block weight × moves ÷ working days); T24 is collecting the inputs
- 97 / 194 is renamed “4-transporter subset (Yim 2008)”, for comparison only; T22 uses 300 / 450 / 600 a day, twice that at peak, and reports both 16-h and 24-h days.

TASKS A DAY (16 H)	FREE FLOW	SEGMENT REQUEST (REFERENCE RULE, TELEPORT)	WHOLE-ROUTE RESERVATION
194 (4-transporter subset, peak)	10	12	12
300	15	19	22
400	20	27	36
600	30	46–50	101–105
800	40	71–75	> 150 (654 at K = 150)

Approximation: vehicles needed = first K whose saturated throughput reaches 99% of the daily volume (Okpo, mean of 10 seeds); above K = 45 the grid steps by 5, hence ranges. The 194 row matches the simulated “10 against 12”. T22 re-runs these with finite volumes.

Sources: Yim et al. 2008, Table 1 caption and “하루 300 개 이상”; Shen et al. 2018 (author's review and guidance, 9 Oct 2026, §2.1 and appendix B; read in full, to be rechecked against the PDFs before the paper). Table values are computed by this deck's make_data.py from Okpo's saturated curves at thesis-core 83c16bf and match the table of guidance §2.1 cell by cell. Conditions: Okpo rev 1824, each road one resource, 16-h days.

The work-zone chart (preview): near 450 tasks a day Okpo's reservation rule needs K^* vehicles; real fleets have 30–40



30 vs 101–105

600 a day: vehicles needed, free flow against whole-route reservation

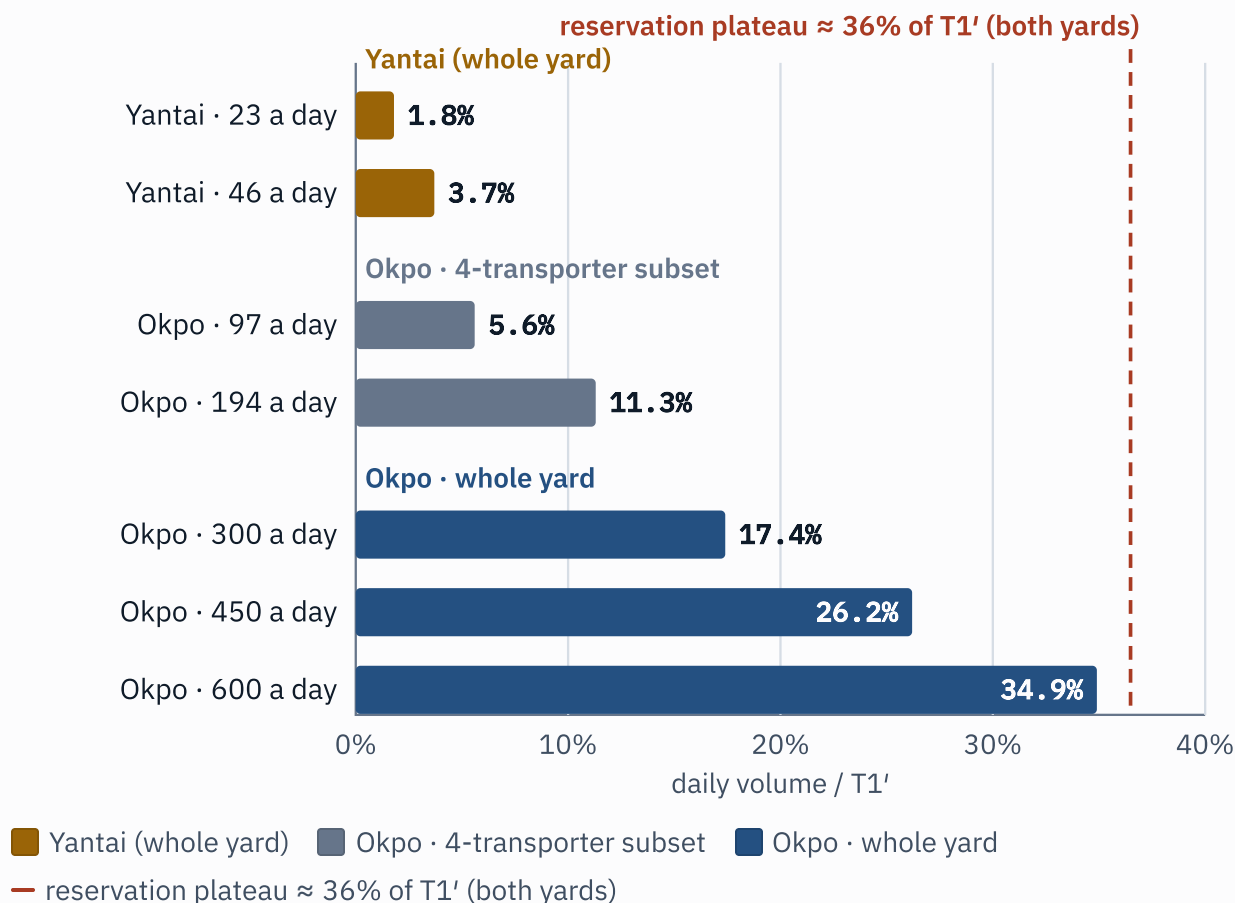
≈ 450

Daily volume at which reservation needs $K^* \approx 50$ vehicles: beyond it, network-bound by the manifesto's vehicles-needed criterion

A tension to answer head-on: the anchor “about 600 a day with about 30 vehicles” matches free flow's 30. The gap between model and anchor is the question; normalise by cycle time first (external anchors).

Source: this deck's make_data.py, reading Okpo's saturated curves at thesis-core 83c16bf with “throughput $\geq 99\% \times$ daily volume”, one point per 25 tasks, at the grid K reached (steps of 5 from $K = 45$). Layout as in the review, §2.1 (daily volume across, vehicles needed up, one line per rule, a band for real fleets, a line for K^*). Real fleets: about 30 in Shen et al. 2018; 33–40 at Hyundai Heavy from press reports, to be checked. T22 draws the formal chart.

The density criterion puts the two yards on the two sides of H1: Yantai's daily volume is 2–4% of T1' , Okpo's whole-yard volume up to 35%



Yantai: the side where free flow suffices

23–46 a day (whole-yard, from public steel use): 1–2 vehicles under every rule, far below $K^* \approx 25$.

Okpo: the side near the network-bound regime

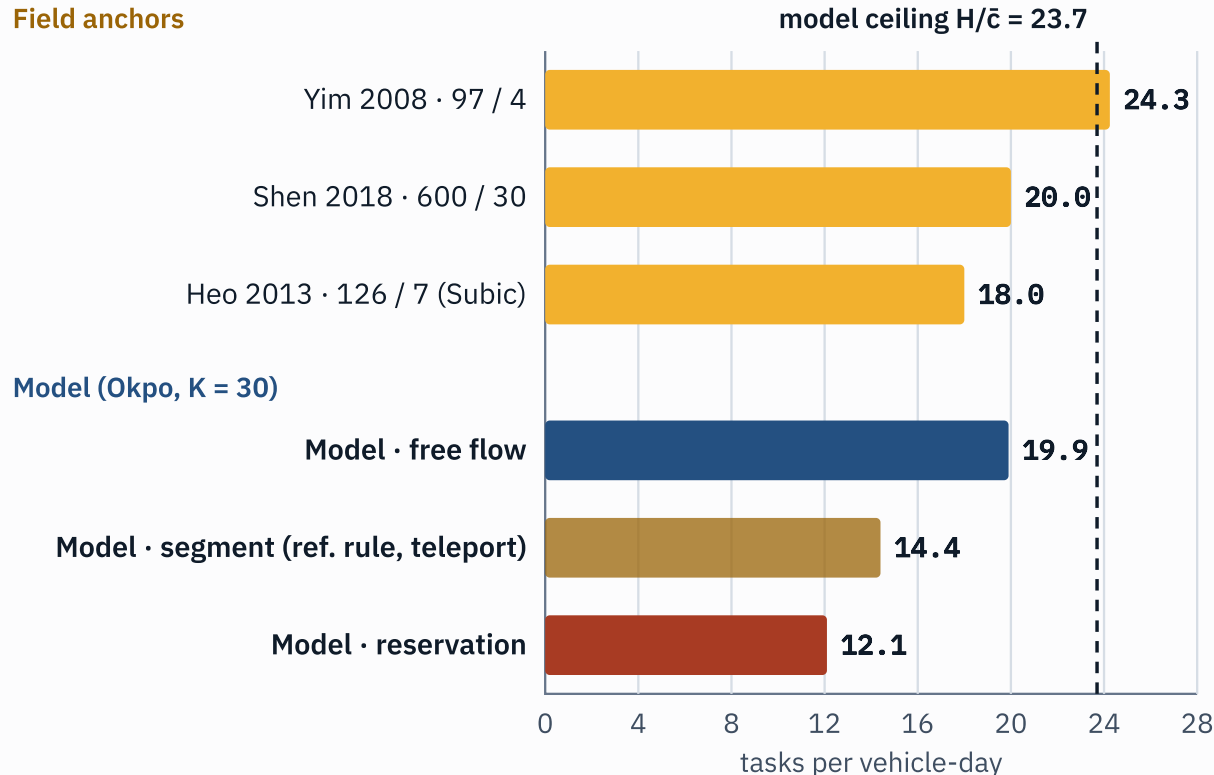
300–600 a day at whole-yard scale is 17–35% of T1'; 600 is already 96% of the reservation plateau of 627, and the vehicles needed pass K^* (preview).

Yantai's T1' binds at the two doors of building 014 (doors held exclusively while loading). The author's imagery check: vehicles can pass in front, so this granularity is conservative; T25 adds a door-granularity scenario. Key-element review is now a fixed step: after T1' for each yard, the author checks the top-3 dual-price elements on the imagery and the resource granularity follows.

Sources: this deck's make_data.py (daily volume $\div$ T1'; T1' and vehicles needed from thesis-core 83c16bf, Okpo a0ab16fceb94, Yantai b82aa19b22af); Yantai's volume in the T20 report §3; binding element, imagery check and the key-element review step in 00_总控/tasks.md (9 Oct 2026) and 船厂清单.md §3 item 4. Both yards' reservation plateaus are 36% of T1'. Conditions: whole-route reservation, each road one resource; Okpo rev 1824, Yantai rev 395; 16-h days.

External anchors: 18–24 blocks per vehicle-day in the field, 19.9 in model free flow, 12–14 under occupancy; normalise first

Field anchors



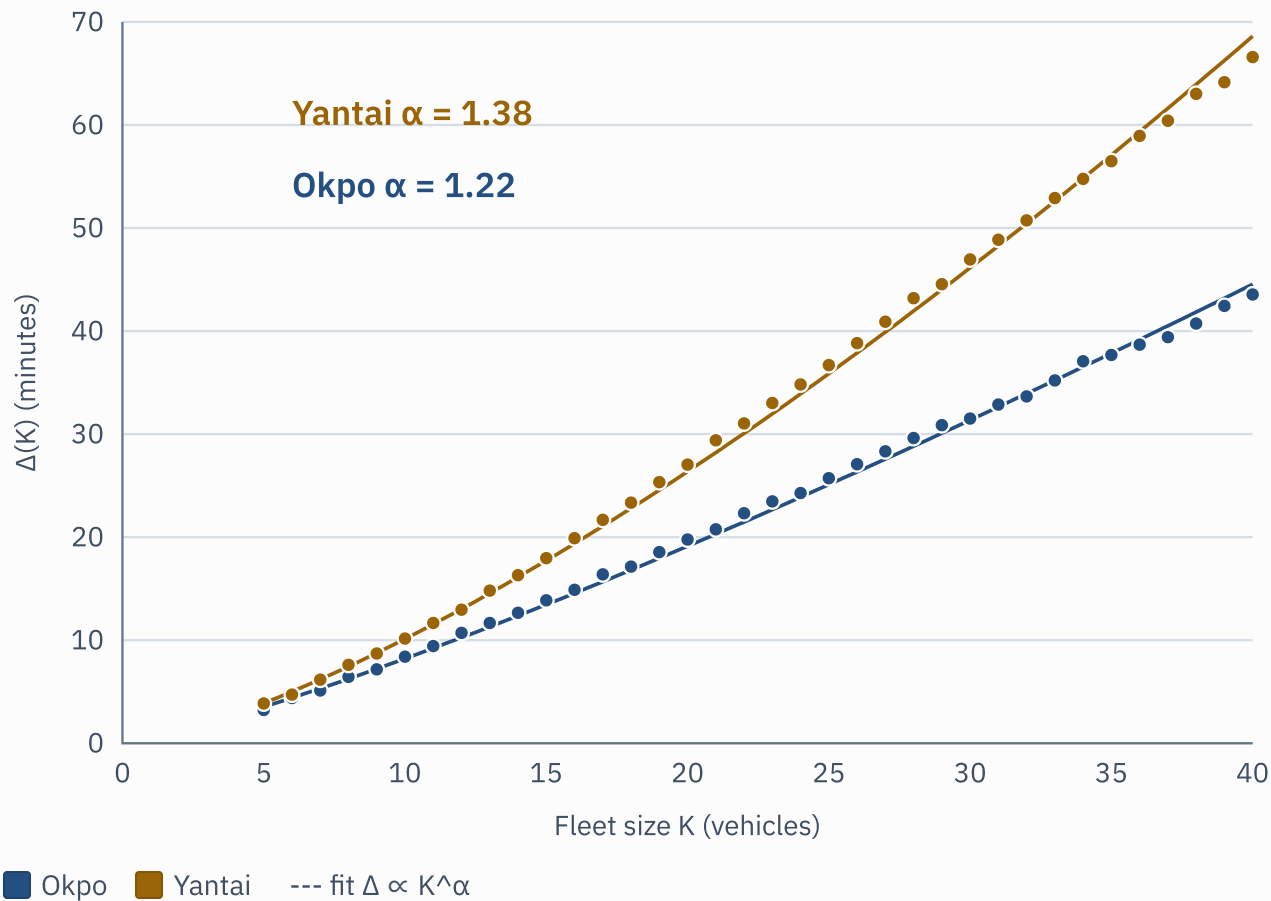
Field anchors Free flow Whole-route reservation

Segment request (reference rule, teleport)

- **Normalise:** per-vehicle output is set mainly by cycle time (distance, handling). The model's ceiling $H/\bar{c} \approx 23.7$ a day ($\bar{c} \approx 40.5$ min) is below Yim's 24.25, so the anchor yard's cycle is shorter; free flow matching the anchors may partly be chance
- **If a gap remains, both explanations go into the paper:** the model is conservative (whole long roads held, empties occupying, 16 h against 24 h, the block-width distribution) → calibrate; the field orchestrates better (signalmen, right of way, one-way loops) → chapter 4's target
- **Road load:** Song & Kang 2009 report 102–161 passes a day on main roads and segment occupancy up to about 26%; to be compared with the model's per-road passes and busy shares at the same volume (whether their yard is Okpo is to be checked)

Sources: anchors 97 / 4 (Yim et al. 2008), 600 / 30 (Shen et al. 2018), 126 / 7 (Heo et al. 2013, Subic, 5 main and 2 auxiliary transporters), the normalisation, both explanations and the road load from the review, §2.1 and §3 Q1, read by make_data.py from that file; no anchor states its shifts. Model: Okpo saturated throughput at $K = 30 \div 30$ (thesis-core 83c16bf); $\bar{c}$ is the mean of 10 seeds in bounds.csv, $H = 960$ min. Conditions: Okpo rev 1824, each road one resource; segment request is the reference rule. T24 is checking these anchors.

H3, the free-flow error: the service-time gap grows superlinearly with fleet size, $\alpha = 1.22$ (Okpo) and 1.38 (Yantai), both above 1



Manifesto criterion: service-time gap $\Delta(K) \propto K^\alpha$ with $\alpha > 1$; the ratio of vehicles needed is read at whole-yard volume.

44 / 67 min

Extra mean service time (dispatch to unloaded) of whole-route reservation over free flow at $K = 40$: Okpo / Yantai

TASKS A DAY	FREE FLOW	WHOLE-ROUTE RESERVATION	RESERVATION ÷ FREE FLOW
194	10	12	1.2
300	15	22	1.5
400	20	36	1.8
600	30	101–105	3.4–3.5

The ratio of vehicles needed grows with daily volume (Okpo preview): the denser the yard, the more free flow underestimates. T22 gives the formal readings at whole-yard volume.

Sources: mean service time (saturated) in curves.csv of each batch at thesis-core 83c16bf, Δ = reservation – free flow; α is the log-log least-squares slope over $K = 5\text{--}40$ (as in experiments/t17_flag.py, line 595). Okpo's 1.22 matches summary.json; fitted over all K (2–150), Yantai gives 1.29 (summary.json). Ratios from the preview table. Conditions: whole-route reservation (just in time), saturated demand, each road one resource; Okpo rev 1824, Yantai rev 395.

What is established, and what is still being settled, each with the task that settles it

Established (each road one resource; Okpo rev 1824, Yantai rev 395)

- **T1' bounds the occupancy models:** the proof draft covers reservation and any rule meeting (A1)–(A3); checked case by case, none above (17,760 runs on both yards in T21)
- **Capacity interval:** Okpo [627, 1,720], Yantai [455, 1,252]; the floor is 36% of the ceiling on both (saturated, whole-route reservation)
- **First two parts of C2:** on both yards a linear rise, then a plateau independent of fleet size
- **H1, part 1:** $K^* \approx 50$ (Okpo), ≈ 25 (Yantai); Yantai sits on the side where free flow suffices
- **H3, first criterion:** $\alpha = 1.22$ and $1.38 > 1$ ($K = 5\text{--}40$)

Still being settled (running or queued, not results)

- **H1 part 2 and H3's vehicle ratio:** whole-yard finite volumes, 16 h and 24 h $\rightarrow$ T22
- **The floor:** whether the safe-release rule segment_safe (Banker-type deadlock avoidance, no teleport) lifts it $\rightarrow$ T22
- **The ceiling:** a system T1' without sampling noise, the route-free T1" $\rightarrow$ T23
- **H1 part 3:** the dock mouth under physical rules and a segmented dock road $\rightarrow$ T22; the three-resource small network $\rightarrow$ T25
- **Okpo's own daily volume and the external anchors** $\rightarrow$ T24

Next: five tasks, and the figure, table or section of the thesis each one feeds

TASK	CONTENT	OUTPUT (PLACE IN THE THESIS)	STATUS
T22	Whole-yard volumes (300 / 450 / 600 a day, peak ×2, 16 h and 24 h); the safe-release rule segment_safe and a no-clearing switch; dock road in sections with direction locks; crane starvation; both yards' p curves	Chapter 3 work-zone chart; new numbers for H1 part 2 and the note on part 3; the interval's floor; per-vehicle comparison	running
T23	System T1' without sampling noise; route-free T1'' and the value of route choice; top-10 dual prices	Chapter 3 bounds section and the ceiling numbers in the text; input to chapter 5's exchange-rate table	running
T24	External anchors and citation checks (Yim, Shen, Heo, whether Song & Kang is Okpo, fleet-size reports); inputs to convert Okpo's daily volume	Chapter 3 external comparison table; one volume caliber for both yards	running
T25	Upgrade simulations and the first road–vehicle exchange-rate table; capacity against block length; the three-resource dock network; Yantai's door granularity	Chapter 5 exchange-rate table; the C3 / C4 boundary; Yantai's T1' at door granularity	queued
T26	Okpo on rev 1833 (floating docks and branch roads) with a floating-dock scenario; finite holding area S and K ₀ (S)	Okpo's final instance; C2's falling part (H2); the H7 case	queued

Sources: 00_总控/tasks.md, rows T22–T26 and the thesis-direction section (9 Oct 2026); the review, §4.1 (T22 order and outputs). T22, T23 and T24 run in parallel; T25 and T26 follow.

Risks and mitigations: each has a way through

RISK	MITIGATION
At whole-yard volume, the occupancy models need far more vehicles than the roughly 30 in the field	Normalise by cycle time first; if a gap remains, write both explanations: calibrate the model (segmented blocking, shifts, width distribution) or make it chapter 4's orchestration target. Either way, “does free flow suffice” gets a real magnitude
Under physical rules and segmented blocking, Okpo's dock-mouth fall weakens or vanishes	H1 part 3 is already a conditional reading (where vehicles wait, how the road is sectioned) plus a design lever, not one yard's threshold; group-A yards with narrower, longer dock roads add cases
Competition: Tao et al. 2017 simulated “road interference” in eM-Plant and adjusted the fleet (abstract only, full text to be checked); Chang's group (Sun 2024) may continue	Get the full text and separate point by point in the introduction: this work formalises exclusive resources and blocking time, compares rules and deadlock, gives a bound and covers two or more yards; no 2025–2026 follow-up from Sun's group was found, and submission is planned within 3–4 months
Only two yards so far	Compute T1' first for the six group-A yards (a few seconds of LP each) and simulate after; chapter 6 picks yards by key-element type

Sources: the review, §5 (risks and mitigations) and §3 Q6; Tao et al. 2017 is abstract-level evidence and, under the research principles, is only marked “full text to be checked”, not treated as pre-emption.

The first paper: working title and candidate journals (chosen at submission, by the results)

When transporters own the road: capacity bounds and operating regimes in shipyard block logistics

Working title; “and dock-mouth instability” may be added if the dock proposition holds

- **Model and T1’**: a rule-independent bound with width bin-packing and crane rows
- **Capacity interval and work zones**: the density and “vehicles needed $\geq K^*$ ” criteria separate the two yards
- **Dock mouth**: where vehicles wait and how the dock road is sectioned (settled by T22)
- **The C3 / C4 boundary and the first road–vehicle exchange-rate table** (T25)

First choice

Transportation Research Part E · IISE Transactions

Capacity and sizing of logistics systems, with models, bounds and simulation together; current results plus the three corrections suffice

With stronger theory

Transportation Research Part B · Transportation Science

Needs two more: a periodic rule reaching $1 - \epsilon$ of T1’ when the dock dominates; a proof of the dock proposition

Alternatives

Transportation Research Part C · European Journal of Operational Research

TR-C if the paper leans to control and orchestration; EJOR if it leans to the LP bound and OR method

Applied

Ocean Engineering

Naval and ocean engineering readers; weight on yard cases, upgrade advice and the toolchain

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Data and files: results/a0ab16fceb94, 49d2383872e7, b82aa19b22af, t18_report_d10b9015cd4d, t19_compare_c963b53c1dde, t20_report_c12289dfda9d and docs/T17–T21 reports at thesis-core commit 83c16bf; research manifesto v1 (tag manifesto-v1); the author's review and guidance (9 Oct 2026); 00_总控/tasks.md. Maps traced by the author on public imagery; this briefing contains neither imagery nor network geometry.

What this sub-study adds over the closest work in the field, and where to go next

1

A bound that holds for any rule

T1' writes occupancy as a structural capacity constraint. Roh & Cha 2011 treat interference as a soft term in the objective; in the search, no comparable bound appears in the shipyard literature.

2

Capacity as a measurable interval

Measured on two real yards as [best deadlock-free rule, T1'], with the floor at 36% of the ceiling: how much orchestration can win back now has a measurable size.

3

Regime criteria that work both ways

K^* , vehicles needed and the density criterion sort yards into “network-bound” and “free flow suffices”, with Okpo and Yantai one example each.

← Thesis briefing

Ceiling: network bound T1' →

Floor: traffic orchestration →

H1 part 3: the dock mouth →

Fleet sizing (MY-B) →

Capacity is not one number but an interval: the traffic rule sets its floor, the network its ceiling.

Thank you.